

- 6) Draw the Straight Line Bode Plot Approximation (magnitude only) for the following transfer function:

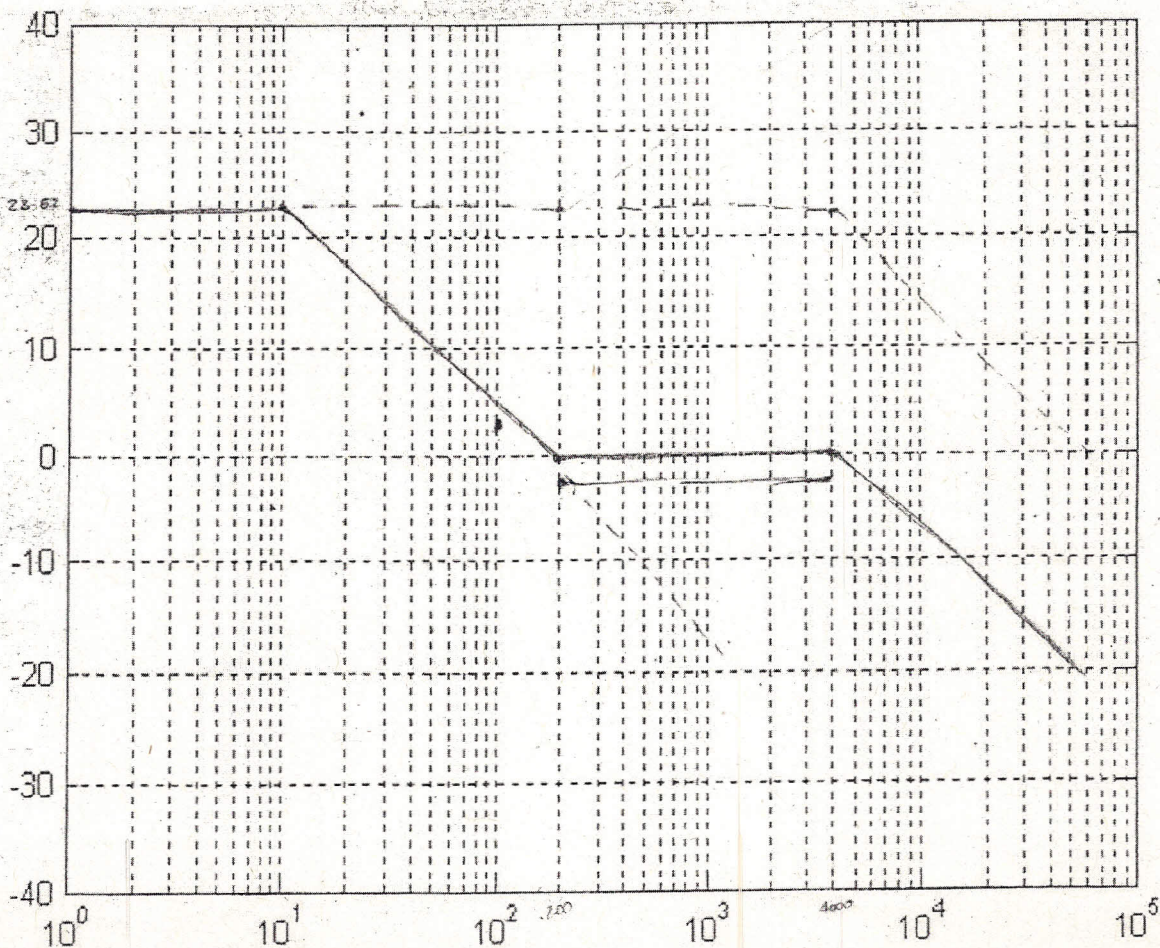
$$H(s) = \frac{3,000 * (s + 200)}{(s + 10) * (s + 4,000)}$$

SHOW all the critical points on your graph.

$$H(s) = \frac{3000 \times 200 \left(1 + \frac{s}{200}\right)}{10 \left(1 + \frac{s}{10}\right) 4000 \left(1 + \frac{s}{4000}\right)} = \frac{15 \left(1 + \frac{j\omega}{200}\right)}{\left(1 + \frac{j\omega}{10}\right) \left(1 + \frac{j\omega}{4000}\right)}$$

$$20 \log 15 = 23.52$$

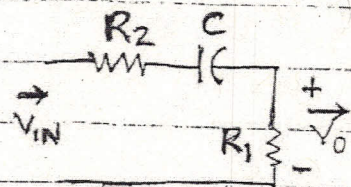
-1



#3)

a) Derive the transfer function

$$H(j\omega) = \frac{\vec{V}_o}{\vec{V}_{in}}, \text{ for the shown circuit}$$



b) Manipulate $H(j\omega)$ to get it into

the following form: $H(j\omega) = \frac{R_1}{R_1 + R_2} \cdot \frac{1}{1 + j\frac{\omega}{\omega_B}}$

Identify the break frequency ω_B .

c) Identify the Imaginary part of $H(j\omega)$.

SHOW YOUR WORK!

a) $-V_{in} + R_2 I + \frac{1}{sC} I + R_1 I$

$V_o = R_1 I$

$-V_{in} + I(R_2 + \frac{1}{sC} + R_1) = 0$

$I = \frac{V_o}{R_1}$

$\frac{V_o}{R_1} (R_2 + \frac{1}{sC} + R_1) = V_{in}$

$$\frac{V_o}{V_{in}} = \frac{R_1}{R_2 + \frac{1}{sC} + R_1} = \frac{R_1 sC}{R_2 sC + 1 + R_1 sC} = \frac{R_1 sC}{1 + R_1 sC + R_2 sC}$$

$H(j\omega) = \frac{j\omega R_1 C}{1 + j\omega(R_1 C + R_2 C)}$

b) $\frac{R_1}{R_1 + R_2} \cdot \left(\frac{j\omega C}{\frac{1}{R_1 + R_2} + j\omega C} \right) \rightarrow \left(\frac{R_1}{R_1 + R_2} \right) \left(\frac{1}{1 + \frac{j\omega C}{R_1 + R_2}} \right)$

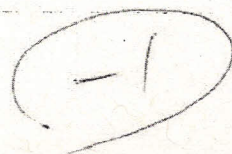
$H(j\omega) = \frac{R_1}{R_1 + R_2} \cdot \frac{1}{1 + j\frac{\omega}{\omega_B}}$

$\omega_B = \frac{1}{C(R_1 + R_2)}$

c) $\frac{1}{1 - j\frac{\omega_B}{\omega}} \cdot \frac{1 + j\frac{\omega_B}{\omega}}{1 + j\frac{\omega_B}{\omega}} = \frac{1 + j\frac{\omega_B}{\omega}}{1 + \frac{\omega_B^2}{\omega^2}}$

Imaginary part

$\frac{j\omega_B/\omega}{1 + \omega_B^2/\omega^2}$



2A) Determine a solution for x and y if

$$\frac{3x - jy}{-2 + j4} = -0.9 + j4.2 = -\frac{9}{10} + j\frac{42}{10}$$

SHOW YOUR WORK

$$\frac{3x - jy}{-2 + j4} = -0.9 + j4.2$$

$$3x - jy = (-0.9 + j4.2)(-2 + j4)$$

$$3x - jy = 1.8 - 3.6j - 8.4j - 16.8$$

$$3x = 1.8 - 16.8$$

$$-y = -3.6 - 8.4$$

$$x = \frac{-15}{3} = -5$$

$$y = 12$$

* 2B) EXPRESS the transfer function

(shown on the right) in
POLAR COORDINATES!

$$H(jx) = \frac{jx / jx}{1 + j3x + (jx)^2 / jx}$$

Identify MAGNITUDE and PHASE Shift!

(NOT DRAW or PLOT)

$$\frac{1}{jx} + 3 + jx$$

$$j\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right) + 3$$

Polar coordinates

magnitude

$$\frac{1}{\sqrt{9 + \left(x - \frac{1}{x}\right)^2}} \angle -\tan^{-1} \left[\frac{\left(x - \frac{1}{x}\right)}{3} \right]$$

phase
shift